

Personal Research Ontologies for Personal Open Science

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Keynote at HCOMP 2017

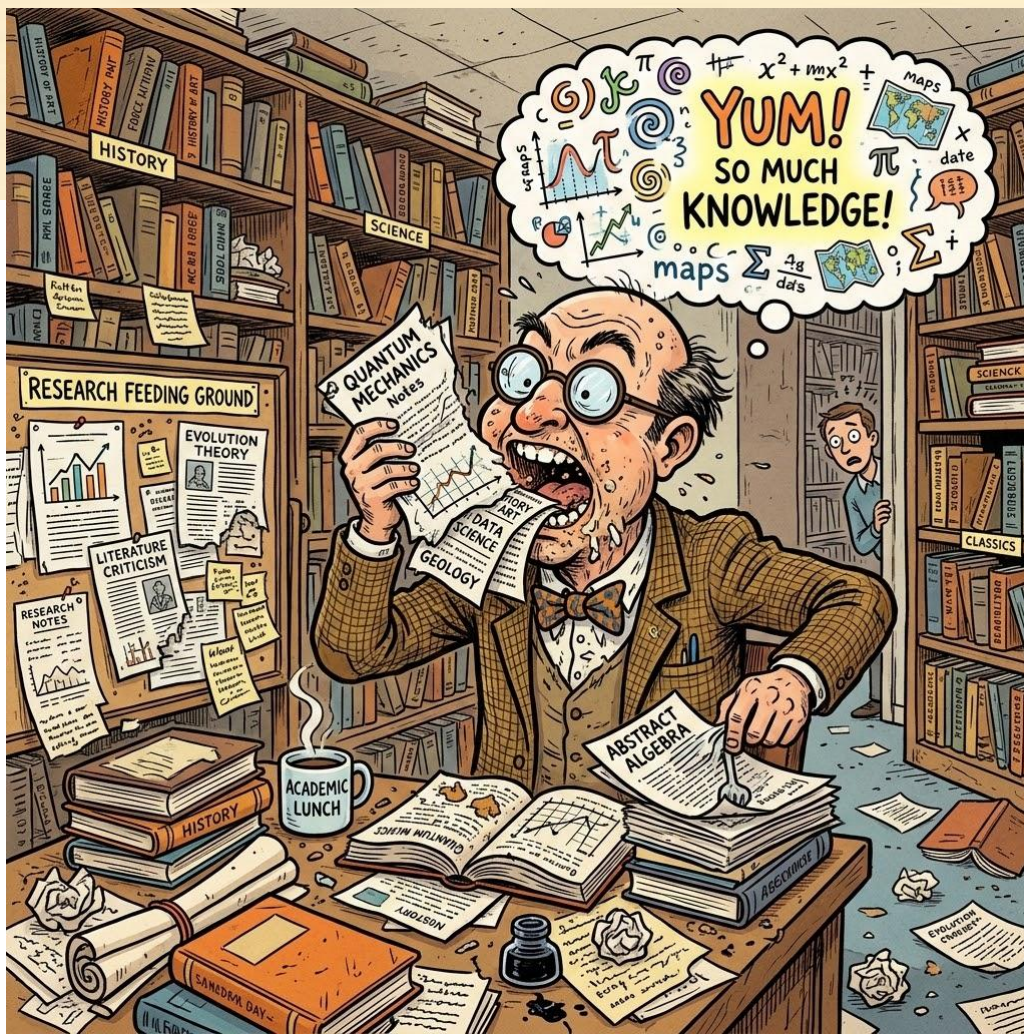


SPEAKER: NIKI KITTUR

Carnegie Mellon University

TOWARDS A UNIVERSAL KNOWLEDGE ACCELERATOR

WEDNESDAY, OCTOBER 25, 2017



Informavore:
An organism that
consumes
information

<https://en.wikipedia.org/wiki/Informavore>

Personal Research Ontologies



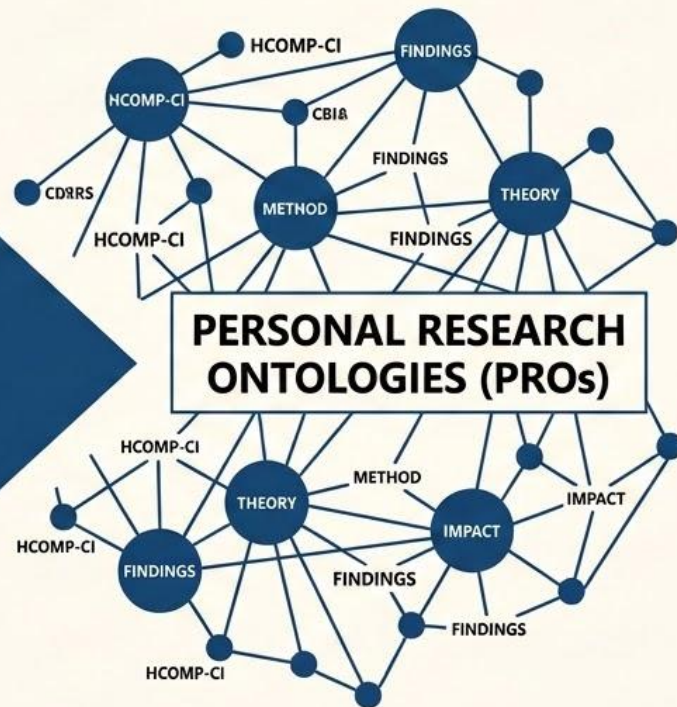
THE RESEARCHER



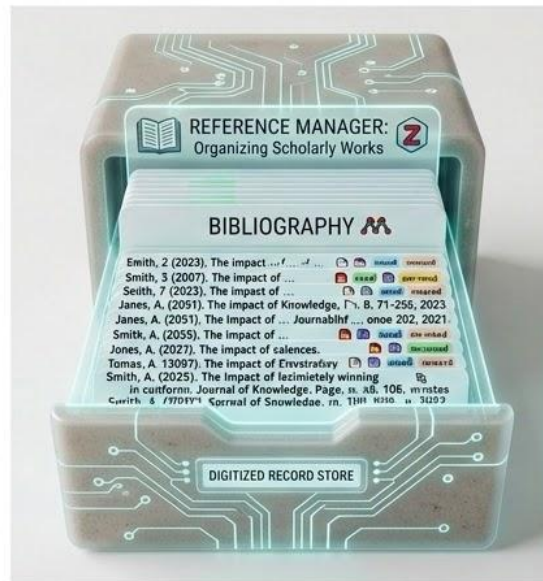
INPUTS
(e.g., Papers, Data, Observations)

RESEARCH IS THE PROCESS OF PERSONAL ONTOLOGY CONSTRUCTION

(The intentional process of conceptualizing, organizing, and defining knowledge structures)



Managing the Personal Research Ontology



Managing the Personal Research Ontology

1. Integration into the daily workflow
2. Low effort and friction
3. Value added
4. Support for exploration and self-reflection

Personal Open Science (Method)

Prefix Meaning

(none)	not classified
-	not relevant/important
!	relevant/important
!!	more relevant/important
!!!	very relevant/important

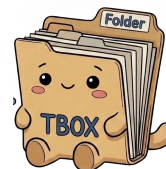
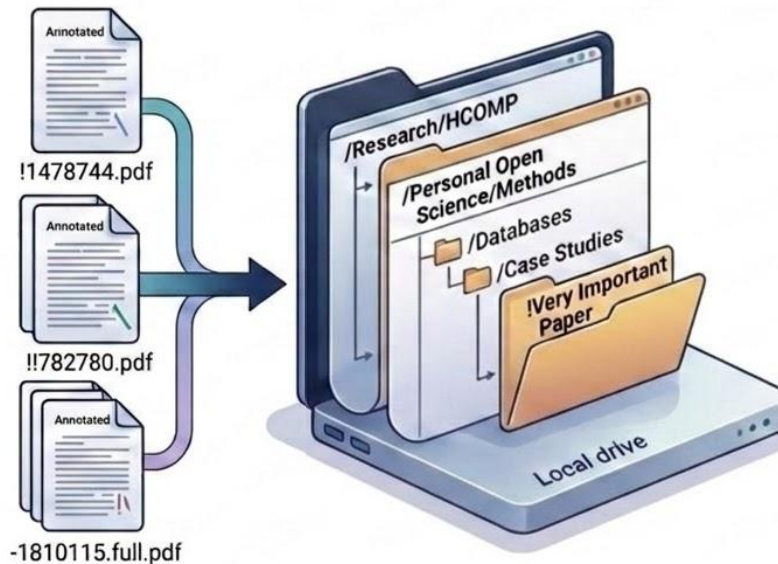


Issues



Applications

...



Downstream Uses (Individual)



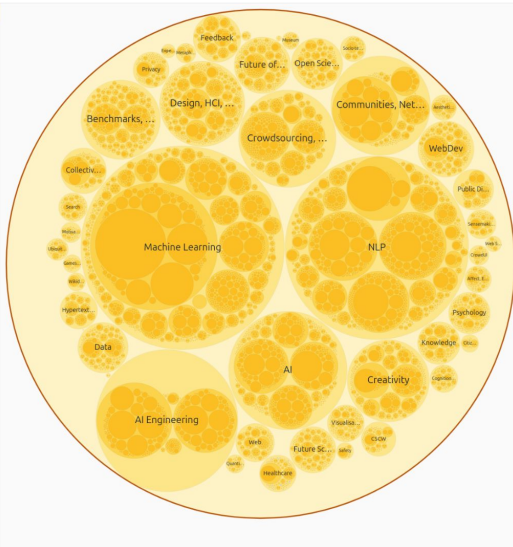
jonaso's research ontology Literatur:Misc · 18,835 folders · 38,189 PDFs · collected May 29, 2026

Concepts Issues

important only

filter folders...

Literatur	30,772
AI	2,348
AI Engineering	1,496
Aesthetics	138
Affect, Emotion	137
Benchmarks, gold standards, datasets	758
CSCW	168
Citizen Science	38
Cognition, Intelligence	171
Collective Intelligence	368
Communities, Networks	1,262
Creativity	1,232
CrowdUI	97
Crowdsourcing, Human Computation	1,985
Data	497
Design, HCI, UI, UX	1,551
Experimentation	27
Feedback	333
Future Science	389
Future of Work	781
Games with a Purpose, GWAP, Gamification	105
Healthcare	245
Hypertext, Hypermedia	258
Knowledge	556
Machine Learning	6,361
Metaphors, Meaning	54
Motivation, Incentives	109
Museum	168
NLP	4,663
Open Science, e-Science, e-Research, Research Data	688
Privacy	238
Search engines	951



You are chatting with 45,444 documents that I have read and found interesting.

restart

What are topics in Collective Intelligence?

Collective Adaptation, Social Computing, Wisdom of the Crowd, Systems Intelligence, Process, Swarms, Applications, Quorum Systems, Comparisons, Ethics, Bees, Collective Computing, Collective Behavior, Trusted AI, Frameworks, Collective Action, Global Challenges, Coordinating Collaboration, Crowd Modelling, Social Choice Theory, Incentives, Global Brain, Wikipedia, Groups, Collective Hearing, Misc, G Factor.

List some prompt engineering techniques.

Scratchpad, Sequential Prompting, Graph of Thoughts, Prompt Dilution, Moral Self-Correction, Active Prompting, Self-Ask Prompt, Chain of Verification (CoVe), Self-refine, Multi-Turn Prompting, Visual Referring Prompting, Prompting with Programs, Multi-Stage Prompting (MSP), Set-of-Mark Prompting, Recursion of Thought, Symbolic Reasoning, BLACK, Constrained Prompting, Chain of Hindsight, Inner Monologue, Re-Act, Inception Prompting, Ask Me Anything, Automatic Reasoning and Tool-Use (ART), Meta-Prompt, Explanations, Manifest, Chain of Density (CoD), Prompt Pattern Catalog, Iterative Prompting, Customized Prompts via Language Models (CuPL), Least-to-most Prompting, Generated Knowledge, Chain of Thought (CoT), Step-Back Prompting, Tree of Thoughts (ToT), Simple and Safe Prompt Engineering (SSP), Chaining, Chains, Self-Consistency, Prompt Pretraining, Paraphrasing, Impersonation, Everything of Thoughts, Progressive Learning, Closed-Loop Reasoning, Plan-and-Solve Prompting, Prompt Of Prompts (POP), AgentInstruct, Retrieval-Augmented Prompting, Dual LLM Pattern, PAL, MRKL, Chain-of-Abstraction Reasoning, Language Model Crossover.

What are the difficulties of moderating twitch communities?

What are the challenges of generative deep learning?

List some prompt engineering techniques.

What are systems, libraries, tools for data validation?

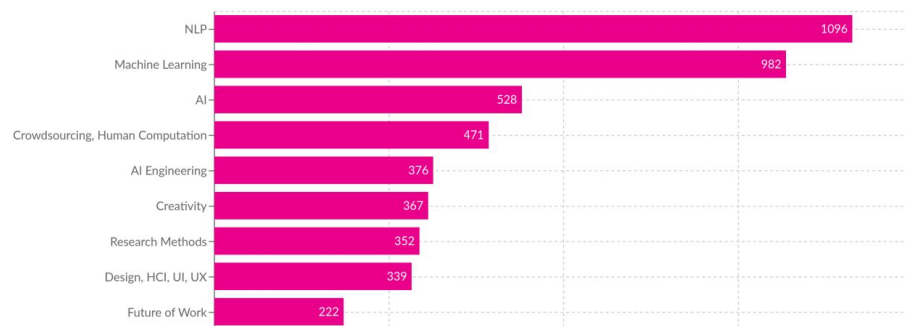
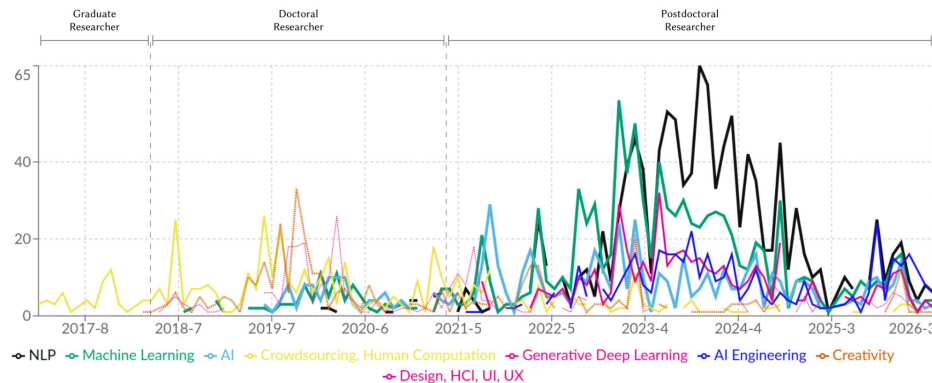
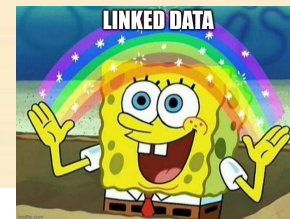
What are the issues with mobile phones?

What are topics in Collective Intelligence?

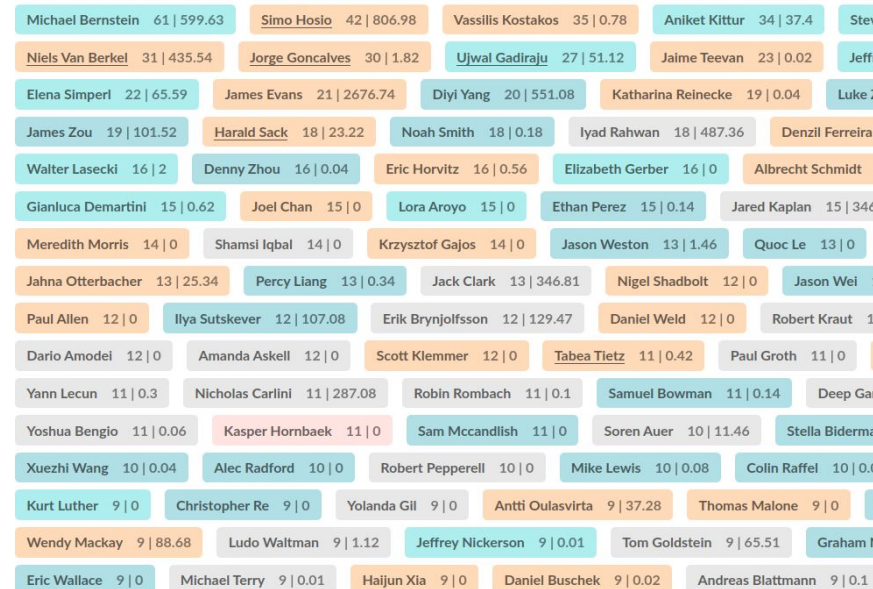
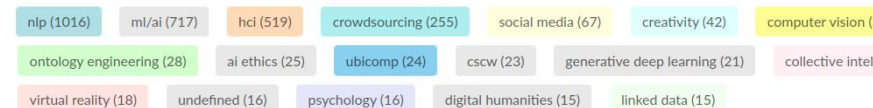
What are the issues with Facebook?



Downstream Uses (Individual)



Research Influences



Personal Open Science (Vision)



INDIVIDUAL FOCUS

Capture more of the
**Personal Research
Ontology**, not just
reading activity



COLLECTIVE IMPACT

Collective benefits
when published
on the Web

Downstream Uses (Collective)



1

**Assessing individual
expertise and learning**



2

**Collaborator discovery
and team formation**



3

**Aggregate field
ontologies**



4

**A new data stream for
the science of science**



5

**A substrate for
distributed and
extended cognition**

Limitations / Future Work



POS METHOD (INDIVIDUAL)

- **Search/retrieval is difficult**
- **Staleness**
(time-based decay?)



POS VISION (COLLECTIVE)

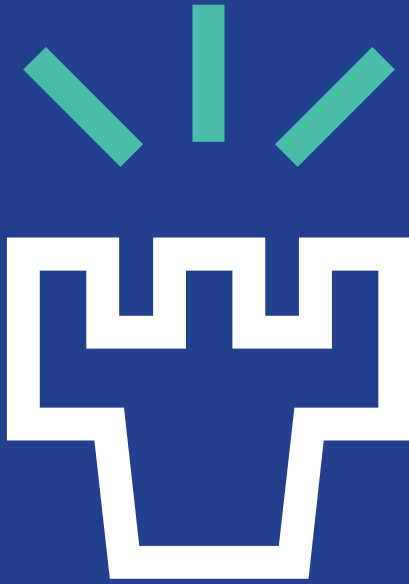
- **Generalization (N=1)**
- **Semantic comparability**
- **Privacy**
- **Gaming**, strategic disclosure
- **Adoption**

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**UNIVERSITY
OF OULU**

PRO for POS

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